

For full credit explain your reasoning, showing all relevant work.

Exercise 1. a) Recall that A_n denotes the subgroup of S_n consisting of all even permutations. Compute all possible orders of elements in A_7 .

b) What is the number of elements of order 6 in A_7 ?

Exercise 2. Let G be a group. Show that the function $f: G \rightarrow G$ defined by $f(a) = a^2$ is a homomorphism if and only if G is an abelian group.

Exercise 3. Describe all possible homomorphisms $f: \mathbb{Q} \rightarrow \mathbb{Z}$, where $\mathbb{Q}$ is the group of rational numbers and $\mathbb{Z}$ is the group of integers (both taken with addition as the group operation). Justify your answer.

PRACTICE PROBLEMS

Exercises below are for practice only - do not turn them in for grading.

Practice Exercise 1. Find an element of order 30 in the group A_{12} and write it in the matrix notation.

Practice Exercise 2. Let H be a subgroup of S_n . Show that if there is $\alpha \in H$ which is an odd permutation, then exactly half elements of H are odd and half are even.

Practice Exercise 3. Let G, H_1, H_2 be groups and let $f_1: G \rightarrow H_1, f_2: G \rightarrow H_2$ be homomorphisms. Show that the function $f: G \rightarrow H_1 \times H_2$ given by $f(a) = (f_1(a), f_2(a))$ is a homomorphism of groups.

Practice Exercise 4. a) Describe all possible group homomorphisms $f: \mathbb{Z}_6 \rightarrow \mathbb{Z}_{15}$.

b) For each of these homomorphisms f compute $\text{Ker}(f)$.